

Vector Potential for the Generalized-Gradient Field Expansion in Frenet–Serret Coordinates with Finite Curvature

1 Introduction

The work of Van der Schueren *et al.* [1] generalizes the generalized gradient formalism of Venturini and Dragt [2] to a curvilinear coordinate system which can have a curvature. This paper describes how the work of Van der Schueren can be applied to do symplectic tracking through bends.

2 Coordinates and Magnetic Field

We work in the Frenet–Serret curvilinear coordinate system (x, y, s) of Van der Schueren [1]. To simplify the analysis, it is assumed that the reference trajectory has constant curvature $h = 1/\rho$ in the x – s plane. This assumption is appropriate for virtually all realistic simulations in high energy storage rings.

Following Van der Schueren, the magnetic field in the curvilinear coordinate system is characterized by a set of functions $a_n(s)$, $b_n(s)$, and $b_s(s)$ where n ranges from one to infinity. Essentially, $n = 1$ is associated with dipole fields, $n = 2$ is associated with quadrupole fields, etc. b_s is associated with solenoid like fields. The association is exact in the limit of $h = 0$ and the functions being independent of s .

The equations of Van der Schueren can be used to construct coefficients that characterize the magnetic field:

$$B_x(x, y, s) = \sum_{i,j=0,0}^{\infty,\infty} C_{x,i,j}(s) x^i y^j \quad (1)$$

$$B_y(x, y, s) = \sum_{i,j=0,0}^{\infty,\infty} C_{y,i,j}(s) x^i y^j \quad (2)$$

$$B_s(x, y, s) = \sum_{i,j=0,0}^{\infty,\infty} C_{s,i,j}(s) x^i y^j. \quad (3)$$

The coefficients C are linear combinations of a_n , b_n , and b_s and their derivatives. Table xxx, which essentially reproduces Table 1 of Van der Schueren, show the lowest order coefficients. Coefficients can easily be calculated using a symbolic math program and stored for use in tracking.

Table 1: Lowest order $C_{x,i,j}$, $C_{y,i,j}$, and $C_{s,i,j}$ coefficients

(i,j)	(0,0)	(1,0)	(0,1)	(2,0)	(1,1)	(0,2)
$C_{x,i,j}$	a_1	a_2	b_2	$\frac{a_3}{2}$	b_3	$\frac{-a_3 - h(a_2 - h a_1 - 2b'_s) - a''_1}{2}$
$C_{y,i,j}$	b_1	b_2	$-b'_s - h a_1 - a_2$	$\frac{b_3}{2}$	$-a_3 - h(a_2 - h a_1 - 2b'_s) - a''_1$	$-\frac{b_3 + h b_2 + b''_1}{2}$
$C_{s,i,j}$	b_s	$a'_1 - h b_s$	b'_1	$h^2 b_s - h a'_1 + \frac{a'_2}{2}$	$b'_2 - h b'_1$	$-\frac{a'_2 + h a'_1 + b''_s}{2}$

3 Vector Potential

For symplectic tracking the vector potential $\mathbf{A}$ is needed. In curvilinear coordinates the the relationship between the vector potential and the field is

$$B_x = (\nabla \times \mathbf{A})_x = \frac{\partial A_s}{\partial y} - \frac{1}{1+hx} \frac{\partial A_y}{\partial s} \quad (4)$$

$$B_y = (\nabla \times \mathbf{A})_y = \frac{1}{1+hx} \frac{\partial A_x}{\partial s} - \frac{1}{1+hx} \frac{\partial}{\partial x}((1+hx) A_s) \quad (5)$$

$$B_s = (\nabla \times \mathbf{A})_s = \frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}. \quad (6)$$

The vector potential is calculated by integrating the above equations along with Eq. (1) through Eq. (3). Evaluation at the integrals lower bounds

Van der Schueren Eqs. (8) through (10) is used to for the evaluation at the integrals lower bounds:

$$B_x(x, y=0, s) = \sum_{n=1}^{\infty} a_n(s) \frac{x^{n-1}}{(n-1)!} \quad (7)$$

$$B_y(x, y=0, s) = \sum_{n=1}^{\infty} b_n(s) \frac{x^{n-1}}{(n-1)!} \quad (8)$$

$$B_s(x=0, y=0, s) = b_s(s). \quad (9)$$

It is convenient to divide the calculation of the vector potential into three pieces by writing

$$C_{x,i,j}(s) = \alpha_{x,i,j}(s) + \beta_{x,i,j}(s) + \gamma_{x,i,j}(s) \quad (10)$$

$$C_{y,i,j}(s) = \alpha_{y,i,j}(s) + \beta_{y,i,j}(s) + \gamma_{y,i,j}(s) \quad (11)$$

$$C_{s,i,j}(s) = \alpha_{s,i,j}(s) + \beta_{s,i,j}(s) + \gamma_{s,i,j}(s) \quad (12)$$

where the α terms are the part of C that depend upon a_n , the β terms are the part of C that depend upon b_n and the γ terms are the part of C that depend upon b_s . For the α piece, the gauge

where $A_y = 0$ is used to produce:

$$A_x = - \sum_{i,j=0,0}^{\infty,\infty} \frac{1}{j+1} \alpha_{s,i,j} x^i y^{j+1} \quad (13)$$

$$A_y = 0 \quad (14)$$

$$A_s = \sum_{i,j=0,0}^{\infty,\infty} \frac{1}{j+1} \alpha_{x,i,j} x^i y^{j+1} \quad (15)$$

For the β part the same gauge $A_y = 0$ is used. Unlike the α piece, the β piece has a nonzero value on the midplane ($B_y(x, 0, s) = \sum_n b_n x^{n-1}/(n-1)!$), so an additional y -independent term, built from the $\beta_{y,i,0}$ coefficients is needed in A_s :

$$A_x = - \sum_{i,j=0,0}^{\infty,\infty} \frac{1}{j+1} \beta_{s,i,j} x^i y^{j+1} \quad (16)$$

$$A_y = 0 \quad (17)$$

$$A_s = \sum_{i,j=0,0}^{\infty,\infty} \frac{1}{j+1} \beta_{x,i,j} x^i y^{j+1} - \frac{1}{1+hx} \sum_{i=0}^{\infty} \beta_{y,i,0} \left(\frac{x^{i+1}}{i+1} + \frac{hx^{i+2}}{i+2} \right) \quad (18)$$

For the γ part, the gauge where $A_z = 0$ is used to produce

$$A_x = (1+hx) \sum_{i,j=0,0}^{\infty,\infty} \left[\int ds \gamma_{y,i,j} \right] x^i y^j \quad (19)$$

$$A_y = -(1+hx) \sum_{i,j=0,0}^{\infty,\infty} \left[\int ds \gamma_{x,i,j} \right] x^i y^j \quad (20)$$

$$A_s = 0 \quad (21)$$

The integrals are trivial since γ is characterized by b_s and its derivatives.

References

- [1] S. Van der Schueren et al. “Magnetic Field Modelling and Symplectic Integration of Magnetic Fields on Curved Reference Frames for Improved Synchrotron Design: First Steps”. In: *Proc. 15th International Particle Accelerator Conference (IPAC'24)*. paper TUPS09. Nashville, TN, USA, May 2024. DOI: 10.18429/JACoW-IPAC2024-TUPS09.
- [2] M. Venturini and A. J. Dragt. “Accurate computation of transfer maps from magnetic field data”. In: *Nucl. Instrum. Methods Phys. Res. A* 427.1-2 (1999), pp. 387–392. DOI: 10.1016/S0168-9002(98)01512-7.